

# End-to-end modeling of kilonova light curves and gravitational waveforms from neutron star mergers for possible constraints on nuclear physics parameters

by

Diego Fernando Hernández Porras

Submitted to the Department of Physics  
in partial fulfillment of the requirements for the degree of

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## ABSTRACT

Neutron stars are compact objects whose hydrostatic equilibrium is determined by the equation of state (EOS) of the dense matter in their interiors, which makes them natural laboratories for nuclear physics at densities inaccessible to terrestrial experiments. Although measurements of neutron-star masses, radii, and tidal deformabilities have constrained the pressure of dense matter, there is still no consensus on the composition of the stellar core, in particular on whether a phase transition to deconfined quark matter takes place. Binary neutron-star mergers (BNSMs) probe even higher densities and temperatures and, as demonstrated by the gravitational-wave event GW170817 and its associated kilonova, are a site of rapid neutron-capture (r-process) nucleosynthesis. In these events, the EOS shapes the hydrodynamic evolution of the merger and may therefore leave fingerprints both in the gravitational waveform and in the mass and initial thermodynamic conditions (e.g., density and entropy) of the ejected material. The subsequent nucleosynthesis in the ejecta, however, also depends on the properties of neutron-rich nuclei far from stability, including nuclear masses,  $\beta^-$ -decay and neutron-capture rates, and fission properties. Because dense-matter and nuclear-physics uncertainties affect the same observables, disentangling them is essential to determine what current and future multimessenger observations can reveal about dense matter. In this work, we perform a joint analysis of the gravitational-wave and electromagnetic signatures of BNSMs to investigate whether a hadron–quark phase transition leaves an imprint on the gravitational-wave signal and on the ejected material, whose mass and composition determine the radioactive heating that powers the kilonova. To this end, we perform general-relativistic hydrodynamic simulations of an equal-mass BNSM with the Einstein Toolkit, using a hadronic EOS, a hybrid EOS that coincides with it below the onset of a first-order hadron–quark phase transition, and an alternative hadronic EOS. Since the inspiral is identical for the hadronic and hybrid models by construction, post-merger differences can be attributed to the phase transition, while the alternative hadronic EOS allows us to assess whether the combined gravitational-wave and electromagnetic signatures can distinguish a phase transition from a change in the stiffness of hadronic matter. We compare the post-merger gravitational-wave signals and the dynamical ejecta, whose thermodynamic evolution we post-process with the nuclear reaction network WinNet to obtain r-process abundances and radioactive heating rates, from which we compute kilonova light curves.

Finally, we quantify the impact of the phase transition relative to the variation of the hadronic EOS, to uncertainties in nuclear-physics inputs such as fission-fragment distributions and  $\beta^-$ -decay rates, and to numerical resolution.

Thesis supervisor: Neelima Govind Kelkar

Title: Professor of Physics

# Acknowledgments

Este trabajo respresenta el final de una de las etapas mas bonitas de mi vida. Llevaré con mucho cariño el recuerdo de todos mis amigos a donde me lleve la motivación.



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# Chapter 1

## Introduction

In this chapter we justify our work based on current open questions.

Is it possible, with future observations, to infer important physics results such as possible transitions to Quark-Gluon plasma? Why quark gluon plasma is of interents for physcisit? As one may imagine, such relevant questions need a acordingly rigourous answers, that of course needs to be verify with observations.

### 1.1 Dense matter and neutron-star equation of state

This section will present and discuss the state of the art regarding the issue with the stability and structure of neutron stars.

### 1.2 Binary neutron-star mergers as multimessenger laboratories

This section will present and discuss the state of the art of the fundamental physics results infered by observations of neutron star mergers.

### 1.3 Open problems: phase transitions and nuclear-physics uncertainties

### 1.4 Objectives and research questions

### 1.5 Structure of the thesis



## Chapter 2

# Dense Matter and Neutron-Star Structure

- 2.1 Equation of state of cold,  $\beta$ -equilibrated matter
- 2.2 Hadron–quark phase transitions and hybrid equations of state
- 2.3 Thermal effects in the equation of state
- 2.4 Stellar structure: TOV equations and tidal deformability
- 2.5 Observational constraints



# Chapter 3

## Binary Neutron-Star Mergers

- 3.1 Merger phases and remnant fate
- 3.2 Gravitational waves: inspiral tidal effects and post-merger spectrum
- 3.3 Mass ejection: tidal, shock-driven, and disk-wind ejecta
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## Chapter 4

# Numerical Relativistic Hydrodynamics

- 4.1 3+1 decomposition and evolution formulation
- 4.2 General-relativistic hydrodynamics in conservative form
- 4.3 The Einstein Toolkit: initial data, mesh refinement, atmosphere
- 4.4 Gravitational-wave extraction
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## r-Process Nucleosynthesis

- 5.1 Nuclear statistical equilibrium and freeze-out
- 5.2 The r-process: neutron-to-seed ratio, fission cycling
- 5.3 Reaction-network equations and numerical solution
- 5.4 Nuclear inputs: masses, decay rates, neutron capture, fission
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# Kilonova Light Curves

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6.2 Opacities of r-process matter

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# Chapter 7

## Methods

7.1 Hadronic and hybrid equation-of-state models

7.2 Binary neutron-star simulation setup

7.3 From simulation to trajectories: ejecta extraction and expansion

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7.6 Gravitational-wave analysis and detectability metrics

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.



# Chapter 8

## Results

8.1 Stellar properties of the equation-of-state models

8.2 Numerical accuracy and convergence

8.3 Merger dynamics and remnant evolution

8.4 Gravitational-wave signals

8.5 Dynamical-ejecta properties

8.6 Nucleosynthesis and heating rates

8.7 Kilonova light curves

8.8 Joint GW EM signatures

8.9 Uncertainty budget: EOS, nuclear physics, and resolution

8.10 Detectability with current and future facilities

physical mechanisms that produce the messengers. This is the main part of the chapter, and it is divided into three sections, each of which is further subdivided into subsections and subsubsections.



# Chapter 9

## Discussion

### 9.1 Comparison with previous studies

### 9.2 Limitations and model assumptions

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# Chapter 10

## Conclusions and Outlook

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# Appendix A

## Units and conventions



## Appendix B

### Convergence tests



## Appendix C

### Network and equation-of-state configuration details